

Classification Shifting in the Cash Flow Statement: Evidence from India

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ABSTRACT: Prior evidence on classification shifting suggests that managers misclassify operating expenses as income-decreasing special items in order to inflate core earnings in the income statement (McVay 2006). There is limited empirical evidence (Lee 2012) on the use of this manipulation and misclassification strategy in the cash flow statement. In this paper, we present first large-sample evidence, using an improved methodology, that managers manipulate operating cash flows using different misclassification strategies. Specifically, they shift operating cash outflows to investing and financing cash outflows, and investing and financing cash inflows to operating cash inflows. We focus on an emerging market, India, which is characterized by weak corporate governance and investor protection, and the United States and present evidence that the magnitude of such misclassification is higher for the firms in India. We also show that the Indian firms in financial distress are more likely to manipulate operating cash flows as compared to the financially distressed firms in the United States by engaging in the misclassification of cash flows. Thus, we link weak governance and investor protection with the magnitude of cash flow manipulation.

Keywords: *cash flow manipulation; classification shifting; operating cash flows; distress.*

1. Introduction

The literature on misclassification of items in the financial statements is growing. Managers inflate core or operating earnings by shifting operating expenses to income-decreasing special items in the income statement (McVay 2006, Fan et al. 2010, Cain et al. 2012). Capital market incentives like meeting or beating earnings benchmarks motivate managers to indulge in these manipulation tactics (McVay 2006, Fan et al. 2010, Athanasakou et al. 2011).

Although the biggest accounting frauds we have witnessed primarily involve earnings manipulation, existing evidence also suggests that managers even manipulate operating cash flows. In the United States, several firms (Dynergy, Worldcom, HealthSouth, etc.) have been accused of manipulating cash flows and engaging in the misclassification in the cash flow statement.

The empirical evidence is in agreement with the real life evidence. Zhang (2006) finds that cash flow manipulation is more likely when magnitude of accruals is high; firms are capital intensive, and when firms consider cash dividend target important. Hollie et al. (2011) discuss that firms overstate operating cash flows through classification shifting. Similarly, using a small sample of cash flow restatement firms, Lee (2012) finds that firms are more likely to upward manage operating cash flows through classification shifting in the cash flow statement when these are in financial distress, have long-term credit rating near the investment/non-investment grade cut-off, have analysts' cash flow forecasts, and there is higher association between firm's stock returns and its cash flow from operations. However, there is no evidence yet about the widespread use of this cash flow manipulation tool by the firms in the United States.

In this paper, we extend the literature on classification shifting in the cash flow statement and present large-sample evidence from an emerging market, India, on the methods of classification shifting. Use of Indian data provides an interesting setting as the country is characterized by weak corporate governance and investor protection (Narayanaswamy et al. 2012) and the presence of family firms and controlling shareholders. Further, unlike the United States it is uncommon to see Indian firms being caught proactively for the accounting wrongdoings.

We present evidence using an improved methodology that the managers of Indian firms manipulate operating cash flows by shifting operating cash outflows to investing and financing cash outflows and by shifting investing and financing cash inflows to operating cash inflows. Our findings suggest that on average an Indian firm inflates approximately Indian National Rupees (INR) 260 million (US\$ 4.33 million)¹ worth of operating cash flows every year. We also examine the use of classification shifting in a specific setting – financial distress, and find that the managers of Indian firms in financial distress are more likely to manipulate operating cash flows through misclassification of cash flows.

We also hypothesize and find that the magnitude of such misclassification in the cash flow statement is higher for the firms in India as compared to those in the United States. This is also true for the financially distressed firms in India, which are more likely to misclassify cash flows. Weaker corporate governance and less regulatory oversight seem to be leading us to these findings.

¹ Throughout the paper, we have assumed a currency exchange rate of Indian National Rupees 60 per United States Dollar.

Overall, this paper contributes to the existing literature on classification shifting in many ways. First, we present evidence that managers misclassify cash flows using different strategies. Second, to the best of our knowledge, this is the first study to examine classification shifting in the cash flow statement in an emerging market, India. Finally, we indirectly relate corporate governance and investor protection with the cash flow manipulation and show that such manipulation is higher in the countries with weaker governance (like India).

In the next section, we discuss motivation for this study and our hypotheses. Section 3 details sample selection and descriptive statistics. In Section 4, we discuss our research design. Section 5 contains results and Section 6 concludes the study.

2. Motivation and hypotheses

There is a sizeable literature on earnings management (see Dechow et al. 2010). However, research on classification shifting in the income statement is relatively recent. Managers of firms in the United States shift operating expenses to income-decreasing special items (McVay 2006, Fan et al. 2010, Cain et al. 2012) in order to inflate core earnings. The primary incentives for expense shifting emerge from desire to meet or beat zero and prior-period earnings (Fan et al. 2010), and to meet or beat analysts' forecasts (McVay 2006, Fan et al. 2010, Athanasakou et al. 2011, Lin et al. 2006). Evidence to this effect also exists for East Asian countries (Haw et al. 2011).

On the other hand, research on cash flow manipulation and classification shifting in the cash flow statement is relatively scarce. As more number of firms and analysts now issue cash flow

forecasts (DeFond and Hung 2003, DeFond and Hung 2007, Wasley and Wu 2006, Call 2008), the probability of manipulation of operating cash flows has increased over the years. Investors have also started paying more attention to the cash flow from operations figure (Schilit and Perler 2010), perhaps because of accounting scams by firms like Enron Corporation in the United States and Satyam Computer Services Limited in India. Evidence of positive stock price reaction also exists, when cash flow surprise is positive (DeFond and Hung 2003, Zhang 2007, Brown et al. 2010).

Operating cash flows are considered sustainable and have valuation related consequences. Besides cash flow based debt covenants, and stock and cash flow based compensation also motivate managers to indulge in creative cash flow reporting (Mulford and Comiskey 2005, Frankel et al. 2010). Firms manage operating cash flows upwards when magnitude of total accruals is high, and discretionary accruals management is low. Such firms belong to capital intensive industries, and consider cash dividend target as important (Zhang 2006).

Using a sample of restatement firms, Lee (2012) finds that firms are more likely to manage operating cash flows upwards through classification shifting when these are in financial distress or have long-term credit rating near the investment/non-investment grade cut-off. Also, existence of analysts' cash flow forecasts and higher association between a firm's stock returns and its cash flow from operations provide significant incentives to manipulate cash flows. Hollie et al. (2011) examine 57 cash flow restatement firms and find that restated cash flow from operations is 50% less than the originally reported figure. Also, such manipulation is more prevalent in retail and financial sectors and restatement firms are bigger. None of these papers present large

sample evidence through which we get to know whether such manipulation by the firms in the United States is widespread.

In order to inflate operating cash flows, managers can shift investing cash inflows to operating cash inflows. Nautica Enterprises Inc. took advantage of flexibility in the Generally Accepted Accounting Principles (GAAP) and classified proceeds from sale of 'available-for-sale' securities as operating cash inflows rather than investing cash inflows (Mulford and Comiskey 2005). Since proceeds from sale of only 'trading' securities can be classified as operating cash inflows, the company first changed the classification of securities from 'available-for-sale' to 'trading'.² Enron Corporation once used loan proceeds to purchase treasury securities. Later, it sold these securities, and repaid the loan using sales proceeds. Though, there was no impact on financing and total cash flows, operating (investing) cash flows did increase (decrease) as sale (purchase) of treasury securities was classified as operating cash inflow (investing cash outflow) (Mulford and Comiskey 2005).

Similarly, managers can also shift financing cash inflows to operating cash inflows. Dynergy, Inc. entered into a complex natural gas purchase contract with its unconsolidated subsidiary – ABG Gas Supply, Inc., where subsidiary borrowed \$300mn from CitiGroup, Inc. enabling it to sell gas at below-market rates to Dynergy. Dynergy sold this gas at market rate for 9 months in 2001, thereby temporarily boosting its operating cash flows (Mulford and Comiskey 2005). Asbury Automotive Group, Inc. recorded change in notes payable for vehicle inventory as an

² These securities were classified as 'available-for-sale' in the balance sheet (year ending February 27, 1999). Proceeds from sale of these securities were classified as cash flows from investing activities in the financial year ending on March 4, 2000, but as cash flows from operating activities in the financial year ending on March 3, 2001.

operating activity rather than financing activity despite the fact that inventory was purchased from a manufacturer unaffiliated with the lender (Hollie et al. 2011).

The discussion above suggests that firms can shift inflows within the cash flow statement categories. However, shifting of outflows is also possible. Specifically, operating cash outflows can be misclassified as either investing or financing cash outflows. HealthSouth Corp., Chambers Development Co., Inc. and Worldcom, Inc. have been accused of reporting operating expenses as capital expenditure to boost operating cash flows. e.g. HealthSouth reported expenses paid for sponsorship and newspaper advertisements as a part of Property, Plant and Equipment (Mulford and Comiskey 2005). There is no available example on firms' attempt to shift operating cash outflows to financing cash outflows. However, we still examine the use of this strategy.

Considering the significance of cash flow as a performance metric, existence of strong incentives behind manipulation, and absence of a large-sample study on classification shifting strategies in the cash flow statement, in this paper, we do a comprehensive analysis of such classification shifting using data from an emerging market, India.

As compared to the United States, India has both weaker corporate governance and investor protection. Unlike United States, class-action lawsuits and lawsuits against the auditors are uncommon here. Further, monetary penalties are miniscule and enforcement of the laws is also weak (Narayanaswamy et al. 2012).

Biggest accounting fraud in India involving Satyam Computer Services Limited came to limelight only when it's Chairman – Ramalinga Raju made a disclosure himself about the accounting malpractices in his company (The Financial Express, 2009). Serious Fraud Investigation Office (SFIO) set up by the Union Government of India, to detect and investigate such frauds failed to uncover this fraud on time. Unlike the Securities and Exchange Commission (SEC) in the United States, SFIO is not known for catching firms for accounting wrong-doings and issuing enforcement actions.

Empirical evidence suggests that the firms in countries with weak investor protection are likely to exhibit more earnings management (Leuz et al. 2003, Haw et al. 2004). Also, use of earnings management through classification shifting in the income statement intensifies with the presence of controlling shareholders (Haw et al. 2011). There is no existing empirical evidence on the influence of corporate governance on cash flow manipulation. Considering the general belief that cash flows are less prone to manipulation, regulators' scrutiny is likely to be less in this case. Further, classification shifting in the cash flow statement is a low cost cash flow manipulation tool with no implications on the future cash flows, and it may be tough for auditors to detect it.

Given the dominant role of family firms in the Indian economy, presence of controlling shareholders, weaknesses in corporate governance and investor protection, Indian firms have more opportunities to engage in operating cash flow manipulation as compared to the firms in the United States. Not only that, Indian firms are also likely to have more incentives to manipulate cash flows through misclassification, as analysts do issue cash flow forecasts here that managers may attempt to meet or beat.

DeFond and Hung (2007) report that about 90% of the Indian firms with analysts' earnings forecasts also have analysts' cash flow forecasts. They suggest that the demand for such cash flow information by the investors is due to the poor quality of earnings reported by the firms in the countries with weak investor protection (like India). Due to the increased opportunities and incentives, we expect that the managers of Indian firms misclassify items in the cash flow statement in order to inflate operating cash flows. Unlike previous studies, we focus on the individual methods of shifting and hypothesize:

H1A: Managers of Indian firms shift investing or financing cash inflows to operating cash inflows.

H1B: Managers of Indian firms shift operating cash outflows to investing or financing cash outflows.

Preceding discussion also suggests that the magnitude of such cash flow misclassification in India is likely to be more as compared to that in the United States due to the increased opportunities to do so.³ Hence, we predict:

H2A: Managers of Indian firms shift investing or financing cash inflows to operating cash inflows more than the managers of firms in the United States.

H2B: Managers of Indian firms shift operating cash outflows to investing or financing cash outflows more than the managers of firms in the United States.

³ We do not expect that Indian firms have comparatively more incentives to misclassify cash flows than the firms in the United States as analysts' cash flow forecast are more common in the United States (DeFond and Hung 2007).

Since, financial distress has been identified as an important firm characteristic associated with the cash flow manipulation (Lee 2012), we also examine whether financially distressed firms in India are more likely to manipulate operating cash flows using classification shifting. Managers of distressed firms consider cash flow as an important performance indicator for outsiders (Graham et al. 2005). Further, cash flows help in predicting bankruptcy (Sharma 2001, Ohlson 1980).

Lee (2012) provides evidence to this effect using a small sample of firms in the United States which had to restate their reported operating cash flows. She finds that firms in financial distress engage in classification shifting to inflate operating cash inflows. Hollie et al. (2011) also find that cash flow restatement firms have higher debt ratio than all firms in the Compustat database.

Our third hypothesis thus makes an incremental contribution and probes whether Lee's (2012) findings can be generalized to a much larger sample in a developing country and whether, financially distressed firms are more likely to engage in shifting using all or few of the above-mentioned methods of shifting.

H3A: Managers of financially distressed firms (in India) are more likely to shift investing or financing cash inflows to operating cash inflows.

H3B: Managers of financially distressed firms (in India) are more likely to shift operating cash outflows to investing or financing cash outflows.

As earlier, we also examine whether there is any significant difference in the magnitude of cash flow misclassification by the financially distressed firms in India and the United States. Bankruptcy costs are comparatively lower in India due to weak firm winding up procedures (Narayanaswamy et al. 2012). Lenders possess less powers and it may take years before they can lawfully take possession of the property of the defaulters. Considering this, financially distressed firms may be less compelled to resort to cash flow misclassification as compared to the distressed firms in the United States. However, on the other hand, magnitude of shifting by the distressed firms in India may be higher than that by the distressed firms in the United States due to the governance issues discussed earlier. Hence, we hypothesize (in null form):

H4: There is an insignificant difference in the magnitude of cash flow misclassification between the financially distressed firms in India and the United States.

3. Sample selection and descriptive statistics

Data for Indian firms has been obtained from Centre for Monitoring Indian Economy's (CMIE) Prowess database and spans from financial years ending in March 1995 to March 2011, as data on cash flows is not well populated in Prowess prior to financial year ending in March 1995. Our sample comprises of non-financial Indian firms listed in A and B groups on the Bombay Stock Exchange, with data available for all dependent and independent variables. Variable definitions are given in Table 1. Initial sample consists of 60,038 firm-years (March 1990-March 2011, Table 2). We remove firms which had a change in fiscal-year end to keep the data comparable. We also remove firm-years for which negative sales or assets values are reported. 22,130 firm-years are lost due to unavailability of data for calculating unexpected operating cash flows. We

require at least 10 firms in each industry-year for running industry-year regressions in order to estimate unexpected operating cash flows. We classify industries using 2-digit National Industrial Classification codes. Regressions are run on a sample with 13,305 firm-years. All variables are winsorized at 1% and 99% by fiscal year. Data for the firms in the United States comes from Compustat North America (Fundamentals Annual). There are 76,258 firm-years in this sample for the years 1989-2010.

/Tables 1 and 2/

Table 3 contains the descriptive statistics for the Indian firms. Median sales (*sale*) and assets (*at*) are INR 974.40 million and INR 1077.30 million, respectively.⁴ Compared to Prowess population, our sample comprises of large firms. Median sales and assets for the population are INR 518.90 million and INR 641.40 million, respectively. Mean (median) operating cash flows (*cfo*) of sample firms are 7.1% (7.4%) of total assets. Altman's (2002) Z-Score (*zscore*) has a mean of 5.10 and median of 4.47. The correlations are reported in Table 4. As hypothesized, unexpected operating cash flows (*ue_cfo*) are significantly positively associated with investing (*ivo*) and financing (*fino*) cash outflows, and significantly negatively associated with investing (*ivi*) and financing (*fini*) cash inflows.

/Tables 3 and 4/

⁴ All amounts are in Indian National Rupees. Amounts in US Dollars are about 16 million and 18 million, respectively.

Sample firms from the United States have median sales (assets) of US\$ 8,480 million (US\$ 8166 million). Median operating cash flows (*cfo*) of these firms are 7.3% of total assets, very close to that of Indian firms. Median Altman's (1968, 2000) Z-Score is 3.159 indicating that firms in the United States are less financially distressed on average.⁵

4. Research design

Our research design is similar to that used in McVay (2006). If managers either shift operating cash outflows to investing or financing cash outflows or shift investing or financing cash inflows to operating cash inflows, a firm's reported cash flow from operations will be higher than what is expected. If classification shifting exists, we expect to see a positive association between the unexpected operating cash flows and investing or financing cash outflows and a negative association between the unexpected operating cash flows and investing or financing cash inflows.

We estimate unexpected operating cash flows using the model proposed by Dechow et al. (1998), and used in Roychowdhury (2006) and Lee (2012).⁶ According to this model (Equation 1), a firm's cash flow from operations (CFO) is a function of sales (S) and change in sales (ΔS). All variables are scaled by lagged total assets.

$$CFO_{i,t}/A_{i,t-1} = \beta_0 + \beta_1 (1/A_{i,t-1}) + \beta_2 (S_{i,t}/A_{i,t-1}) + \beta_3 (\Delta S_{i,t}/A_{i,t-1}) + \epsilon_{i,t} \quad (1)$$

⁵ Figures for this sample are untabulated.

⁶ In order to ensure that the model captures a firm's fundamental performance effectively, we estimate unexpected operating cash flows using two alternative models. First one includes a control for previous year's cash flows, while the second one includes a control for current year's core earnings. Results are discussed in the next section.

Specifically, equation one is run for every industry-year with minimum 10 observations. Expected operating cash flows are estimated for each firm-year using estimated coefficients from equation one. Unexpected operating cash flows (UE_CFO) are calculated as the difference between reported operating cash flows and predicted operating cash flows. We test our first and third hypotheses using the following model (Equation 2), which is based on McVay (2006). Regression is run on pooled data using this equation, with industry and time dummies.

$$\begin{aligned} \text{UE_CFO}_{i,t} = & \alpha_0 + \alpha_1 \text{IVO}_{i,t} + \alpha_2 \text{IVI}_{i,t} + \alpha_3 \text{FINO}_{i,t} + \alpha_4 \text{FINI}_{i,t} + \alpha_5 \text{ZSCORE}_{i,t-1} + \\ & \alpha_6 \text{ZSCORE}_{i,t-1} * \text{IVO}_{i,t} + \alpha_7 \text{ZSCORE}_{i,t-1} * \text{IVI}_{i,t} + \alpha_8 \text{ZSCORE}_{i,t-1} * \text{FINO}_{i,t} + \\ & \alpha_9 \text{ZSCORE}_{i,t-1} * \text{FINI}_{i,t} + \alpha_{10} \text{ROA}_{i,t} + \alpha_{11} \text{SIZE}_{i,t} + \alpha_{12} \text{MTB}_{i,t} + \alpha_{13} \text{DACC}_{i,t} + \omega_{i,t} \end{aligned} \quad (2)$$

UE_CFO is unexpected operating cash flows estimated using equation one. Variable IVO represents cash outflows from investing activities, while variable IVI represents cash inflows from investing activities. FINO and FINI are cash outflows and inflows from financing activities, respectively. Positive (negative) and significant coefficients on IVO and FINO (IVI and FINI) will indicate existence of classification shifting.

ZSCORE represents firm's financial health and is estimated using emerging market bankruptcy prediction model of Altman (2002).⁷ A firm's probability of being financially distressed increases with a decline in ZSCORE. We multiply ZSCORE with *minus* one and expect coefficients α_6 and α_8 to be positive and α_7 and α_9 to be negative, if probability of cash flow manipulation through classification shifting is increasing with financial distress.

⁷ For firms in the United States, we estimate this variable based on Altman (1968, 2000).

Following Lee (2012), we also include controls for return on assets (ROA), size (SIZE), market-to-book ratio (MTB) and discretionary accruals (DACC). Operating cash flows are also affected when real actions like reduction in discretionary expenditure, price discounts, overproduction, etc. are used. These actions also affects earnings and hence, controlling for return on assets (ROA) provides a stronger test for examining cash flow manipulation using only classification shifting. Managers may use cash flow manipulation to either mask accruals management or as a substitute for accruals management (Zhang 2006). Thus use of discretionary accruals (DACC), estimated using cross-sectional version of Jones (1991), as an explanatory variable controls for this association. Size (SIZE) and market-to-book (MTB) ratio have been used to control for any variation in the unexpected cash flow from operations emanating due to firm size and growth opportunities.

Approach used in equation two helps in capturing the use of classification shifting by the sample firms on average, which is not possible by examining only the unexpected operating cash flows (UE_CFO) or a sample of restatement firms (as in Lee 2012). It is also possible that unexpected operating cash flows (UE_CFO) contain a part of firms' normal and un-manipulated operating cash flows. This may happen if the predicted cash flows in the equation one do not fully reflect the firms' fundamental operating performance. In such a case, relying on the association between the unexpected operating cash flows (UE_CFO) and investing (IVO and IVI) and financing (FINO and FINI) cash flows is much useful than only on the value of unexpected operating cash flows (UE_CFO).

In equation two, we measure cash flow variables (IVO, IVI, FINO and FINI) based on the signs of the net cash flows reported. However, in this way, a firm with positive net investing cash flows will be counted in while calculating IVI and not IVO despite the possibility of such firm to have reported an investing cash outflow.^{8,9}

To overcome this limitation, we use net cash flows from investing activities (CFI) and net cash flows from financing activities (CFF) in place of variables IVO, IVI, FINO and FINI. In this manner, we infer shifting from operating cash flows to investing or financing cash flows and vice versa without making any inference about whether outflows were shifted or inflows.

If managers shift operating cash flows to investing cash flows or vice versa, we expect to see a negative association between the unexpected operating cash flows (UE_CFO) and net investing cash flows (CFI). Similarly, if they shift operating cash flows to financing cash flows or vice versa, we expect to see a negative association between the unexpected operating cash flows (UE_CFO) and net financing cash flows (CFF). We use equation three to test this prediction.

$$\begin{aligned} \text{UE_CFO}_{i,t} = & \alpha_0 + \alpha_1 \text{CFF}_{i,t} + \alpha_2 \text{CFI}_{i,t} + \alpha_3 \text{ZSCORE}_{i,t-1} + \alpha_4 \text{ZSCORE}_{i,t-1} * \text{CFF}_{i,t} + \\ & \alpha_5 \text{ZSCORE}_{i,t-1} * \text{CFI}_{i,t} + \alpha_6 \text{ROA}_{i,t} + \alpha_7 \text{SIZE}_{i,t} + \alpha_8 \text{MTB}_{i,t} + \alpha_9 \text{DACC}_{i,t} + \delta_{i,t} \end{aligned} \quad (3)$$

As earlier, we multiply ZSCORE with *minus* one and expect coefficients α_4 and α_5 to be negative, if probability of cash flow manipulation through classification shifting is increasing with financial distress.

⁸ We thank one of the reviewers for pointing out this issue.

⁹ The literature on classification shifting in the income statement (McVay 2006, Fan et al. 2010) suffers from the same bias, where negative values of net special items are considered as income-decreasing special items.

We use equations four and five to test our hypotheses two and four, where we examine if the magnitude of cash flow misclassification differs between the firms in India and the United States.

$$\begin{aligned}
UE_CFO_{i,t} = & \alpha_0 + \alpha_1 IVO_{i,t} + \alpha_2 IVI_{i,t} + \alpha_3 FINO_{i,t} + \alpha_4 FINI_{i,t} + \alpha_5 ZSCORE_{i,t-1} + \\
& \alpha_6 ZSCORE_{i,t-1} * IVO_{i,t} + \alpha_7 ZSCORE_{i,t-1} * IVI_{i,t} + \alpha_8 ZSCORE_{i,t-1} * FINO_{i,t} + \alpha_9 ZSCORE_{i,t-1} * FINI_{i,t} \\
& + \alpha_{10} IND_{i,t} + \alpha_{11} IND_{i,t} * IVO_{i,t} + \alpha_{12} IND_{i,t} * IVI_{i,t} + \alpha_{13} IND_{i,t} * FINO_{i,t} + \alpha_{14} IND_{i,t} * FINI_{i,t} + \\
& \alpha_{15} IND_{i,t} * ZSCORE_{i,t-1} * IVO_{i,t} + \alpha_{16} IND_{i,t} * ZSCORE_{i,t-1} * IVI_{i,t} + \alpha_{17} IND_{i,t} * ZSCORE_{i,t-1} * FINO_{i,t} + \\
& \alpha_{18} IND_{i,t} * ZSCORE_{i,t-1} * FINI_{i,t} + \alpha_{19} ROA_{i,t} + \alpha_{20} SIZE_{i,t} + \alpha_{21} MTB_{i,t} + \alpha_{22} DACC_{i,t} + \mu_{i,t} \quad (4)
\end{aligned}$$

$$\begin{aligned}
UE_CFO_{i,t} = & \alpha_0 + \alpha_1 CFF_{i,t} + \alpha_2 CFI_{i,t} + \alpha_3 ZSCORE_{i,t-1} + \alpha_4 ZSCORE_{i,t-1} * CFF_{i,t} + \alpha_5 ZSCORE_{i,t-1} \\
& * CFI_{i,t} + \alpha_6 IND_{i,t} + \alpha_7 IND_{i,t} * CFF_{i,t} + \alpha_8 IND_{i,t} * CFI_{i,t} + \alpha_9 IND_{i,t} * ZSCORE_{i,t-1} * CFF_{i,t} + \alpha_{10} \\
& IND_{i,t} * ZSCORE_{i,t-1} * CFI_{i,t} + \alpha_{11} ROA_{i,t} + \alpha_{12} SIZE_{i,t} + \alpha_{13} MTB_{i,t} + \alpha_{14} DACC_{i,t} + \Omega_{i,t} \quad (5)
\end{aligned}$$

Here IND is an indicator variable equal to 1 if a firm is Indian and 0 if it is from the United States sample.¹⁰ Coefficients of interest in equation four are α_{11} to α_{18} . We expect coefficients α_{11} and α_{13} (α_{12} and α_{14}) to be positive (negative) if firms in India are more likely to misclassify cash flows than those in the United States. We do not have any directional expectation with respect to our hypothesis four, where we test the use of cash flow misclassification by the distressed firms in both the countries. As earlier, we expect coefficients α_7 and α_8 to be negative in equation five.

¹⁰ These tests use samples of firms from both India and the United States, which have come from two different databases. Since our tests use scaled (by total assets) variables, our inferences rely on the ratios rather than the absolute values and thus, our results should be considered reliable.

5. Results

Table 5 contains the mean coefficients and adjusted R-Square for equation one. Signs of the coefficients on *def_sale* and *ch_sale* are as expected. An increase in sales (*def_sale*) is likely to result in an increase in earnings and cash flows, while an increase in sales growth (*ch_sale*) is likely to result in more accruals and less cash flows. Mean adjusted R-Square is less (16%) as compared to previous studies (45% in Roychowdhury 2006 and 39% in Lee 2012).¹¹

/Table 5/

5.1. Classification shifting

Results for tests of our first and third hypotheses are given in Tables 6 and 7. In both the tables, panel A doesn't contain industry and year dummies while panel B does. In Table 6, coefficients on *ivo*, *ivi*, *fino* and *fini* are highly significant in both panels. As hypothesized, coefficients on *ivo* and *fino* are positive, which suggests that the managers of Indian firms shift operating cash outflows to investing and financing outflows resulting in a simultaneous increase in the unexpected operating cash flows, and investing and financing outflows.

/Table 6/

Similarly, as managers shift investing and financing cash inflows to operating cash inflows, a positive shock to operating cash inflows is accompanied by a decline in investing and financing

¹¹ Mean adjusted R-Square is between 22%-26% when we use alternative models discussed in the last section.

cash inflows. We see this hypothesized association in panels A and B of Table 6. Specifically, coefficients on *ivi* and *fini* are negative and statistically significant.

We get similar results when we use equation three. In Table 7, coefficients on *cff* and *cfi* are negative and statistically significant. Negative coefficient on *cff* (*cfi*) suggests that managers shift cash flows between operating and financing (investing) categories in order to inflate operating cash flows. Thus, operating cash flows increase with a decrease in financing or investing cash flows.¹²

/Table 7/

These results imply a mean shifting per firm-year of approximately INR 44 million¹³ (US\$ 0.73 million) between operating and financing cash flows and approximately INR 216 million¹⁴ (US\$ 3.60 million) between operating cash flows and investing cash flows. Higher magnitude of shifting between operating cash flows and investing cash flows is likely due to the fact that managers have opportunity to shift both investing cash inflows to operating cash inflows and operating cash outflows to investing cash outflows, while shifting between operating and financing cash flows is likely to be due to opportunity to shift only financing cash inflows to operating cash inflows. On average, an Indian firm inflates approximately INR 260 million (US\$ 4.33 million) worth of operating cash flows every year.

¹² All the results (for hypotheses one and two) are qualitatively similar when we use unexpected cash flows estimated using the alternative models discussed in the last section.

¹³ Coefficient on *cff* (0.293) in Panel B of Table 7 multiplied by mean net cash flows from financing activities *fcf* (150.554).

¹⁴ Coefficient on *cfi* (0.340) in Panel B of Table 7 multiplied by mean net cash flows from investing activities *icf* (634.496).

5.2. Impact of financial distress

Results in panels A and B of Table 6 indicate that the financially distressed firms are more likely to use classification shifting in order to manipulate operating cash flows. Coefficients on $zscore*ivo$ and $zscore*fini$ have expected signs and are statistically significant. Coefficients on $zscore*ivi$ and $zscore*fino$ also have expected signs but are not statistically significant.

Positive and significant coefficients on $zscore*ivo$ suggest that the managers of distressed firms are more likely to shift operating cash outflows to financing cash outflows.¹⁵ Similarly, negative and significant coefficients on $zscore*fini$ suggest that managers of financially distressed firms are more likely to shift financing cash inflows to operating cash inflows.

We get similar results when we use equation three. In Table 7, coefficients on $zscore*cff$ and $zscore*cfi$ are negative and significant. Negative coefficients on $zscore*cff$ ($zscore*cfi$) suggest that the managers of financially distressed firms are more likely to shift cash flows between operating and financing (investing) categories in order to inflate operating cash flows.¹⁶ These results imply that as a firm becomes financially distressed, on an average it is likely do an additional shifting of approximately INR 0.60 million¹⁷ (US\$ 0.01 million) per year between operating and financing cash flows and approximately INR 3.17 million¹⁸ (US\$ 0.05 million) per year between operating cash flows and investing cash flows.

¹⁵ Note that we multiply $zscore$ with *minus* one, when we run regressions.

¹⁶ When we use unexpected cash flows estimated using the alternative models discussed in the last section, our findings (related to third and fourth hypothesis) from equation three are qualitatively similar. These findings are mixed when we use equation two. Overall, evidence of shifting is still seen.

¹⁷ Coefficient on $zscore*cff$ (0.004) in Panel B of Table 7 *multiplied by* mean net cash flows from financing activities fcf (150.554).

¹⁸ Coefficient on $zscore*cfi$ (0.005) in Panel B of Table 7 *multiplied by* mean net cash flows from investing activities icf (634.496).

5.3. Impact of weak corporate governance and investor protection

Our second hypothesis predicts that the managers of Indian firms misclassify cash flows more than those in the United States. We observe so in Tables 8 and 9. As expected, coefficients on *ind*ivo*, *ind*ivi*, *ind*fino* and *ind*fini* in Table 8 are statistically significant at one percent level and carry expected signs. This finding is also evident in Table 9, where coefficients on *ind*cff* and *ind*cfi* are negative and highly significant. Thus, Indian firms are more likely to engage in cash flow misclassification than the firms in the United States. As discussed earlier, this is likely due to the increased opportunities emanating from weak governance, regulatory oversight and investor protection in India, rather than the increased incentives as analysts' cash flow forecasts are more common in the United States than India (DeFond and Hung 2007).

Results (Tables 8 and 9) with respect to test of our hypothesis four indicate that the financially distressed firms in India are more likely to misclassify cash flows than such firms in the United States. In Table 8, as expected, coefficients on *ind*zscore*ivo* and *ind*zscore*fini* are significantly positive and negative, respectively. This suggests that misclassification of cash flows increases with financial distress more for the Indian firms than the firms in the United States. Remaining two coefficients are statistically insignificant. However, coefficients on *ind*zscore*cff* and *ind*zscore*cfi* are significantly negative in Table 9 and this result supports our hypothesis. Again, weak governance seems to be leading us to this finding.

In Tables 8 and 9, we observe that cash flow classification shifting is also widespread in the United States. Specifically, coefficients on *ivo*, *ivi*, *fino* and *fini* are highly significant and carry expected signs in Table 8. Similarly, coefficients on *cff* and *cfi* are negative and highly

significant in Table 9. However, financial distress doesn't seem to have a major impact on the use of this shifting as indicated by the magnitudes of the coefficients on $zscore*cff$ and $zscore*cfi$. Overall, our results corroborate the findings of Lee (2012) and Hollie et al. (2011).

To summarize, the results suggest that the managers of Indian firms engage in the misclassification of cash flows in order to inflate operating cash flows. Also, financially distressed Indian firms are more likely to manipulate operating cash flows, to portray a better picture of their financial condition to external parties. This cash flow manipulation is likely to be higher for the firms in India in general and for the financially distressed Indian firms in particular, as compared to the firms in the United States.

6. Conclusion

A large portion of the literature on classification shifting has focused on the misclassification in the income statement (McVay 2006, Fan et al. 2010). There is limited evidence on the use of this manipulation strategy in the cash flow statement (Lee 2012).

In this paper, we present large-sample evidence that the managers of Indian firms manipulate operating cash flows using different misclassification strategies. Specifically, they shift operating cash outflows to investing and financing cash outflows, and investing and financing cash inflows to operating cash inflows. We also hypothesize and find that the firms in financial distress are more likely to manipulate operating cash flows by engaging in the misclassification of cash flows.

We also show that the magnitude of such misclassification in the cash flow statement is higher for the firms in India as compared to those in the United States. This is also true for the financially distressed firms in India, which are more likely to misclassify cash flows than such firms in the United States. Weaker corporate governance, investor protection and less regulatory oversight in India seem to be leading us to these findings.

Our findings should be of much interest to investors and regulators in India and other countries. Results indicate that cash flows are as prone to manipulation and misclassification as the earnings. So, even cash flows may not indicate the true fundamental performance of a firm. Regulators and auditors in India and other countries with weak corporate governance and investor protection need to monitor cash flow reporting very closely.

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Table 1
Variable definitions

at	Total assets in year t .
ce	Core or operating earnings. (Operating income before depreciation and amortization _{t} /Total assets _{$t-1$}).
cff	Net cash flow from financing activities in year t , scaled by total assets in year $t-1$.
cfi	Net cash flow from investing activities in year t , scaled by total assets in year $t-1$.
cfo	Net cash flow from operating activities in year t , scaled by total assets in year $t-1$.
ch_sale	Growth in sales. ((Net sales _{t} -Net sales _{$t-1$})/Total assets _{$t-1$}).
dacc	Discretionary accruals estimated using Jones (1991).
def_sale	Net sales in year t , scaled by total assets in year $t-1$.
fcf	Net cash flow from financing activities in year t .
fini	If $cff > 0$ then $fini = cff$, 0 otherwise.
fino	If $cff < 0$ then $fino = -1*cff$, 0 otherwise.
icf	Net cash flow from investing activities in year t .
ind	Indicator variable equal to 1 if a firm is Indian and 0 if it is from the United States sample.
inv_at	Inverse of total assets in year $t-1$. (1/Total assets _{$t-1$}).
ivi	If $cfi > 0$ then $ivi = cfi$, 0 otherwise.
ivo	If $cfi < 0$ then $ivo = -1*cfi$, 0 otherwise.
mtb	Market-to-book ratio. ((Common shares outstanding _{t} *Fiscal year-end closing stock price _{t})/Common Equity _{t}).
ocf	Net cash flow from operating activities in year t .
roa	Profit after tax in year t , scaled by total assets in year $t-1$.
sale	Net sales in year t .
size	Natural logarithm of total assets in year t .
ue_cfo	Unexpected operating cash flows estimated using equation 1 or 4.
zscore	Altman's (2002) Z-Score in year $t-1$. ((6.56*(Working capital _{$t-1$} /Total assets _{$t-1$})))+(3.26*(Retained earnings _{$t-1$} /Total assets _{$t-1$})) +(6.72*(Profit before interest and tax _{$t-1$} /Total assets _{$t-1$})))+(1.05*(Common equity _{$t-1$} /Total liabilities _{$t-1$}))).

Table 2
Sample selection

	No. of firms	No. of firm-years
Initial Prowess sample with non-missing company code or National Industrial Classification code (March 1990-March 2011)	2,729	60,038
<u>Less:</u> Firm-years of firms with a change in fiscal-year end	760	16,720
	1,969	43,318
<u>Less:</u> Firm-years with negative sales or assets	0	695
	1,969	42,623
<u>Less:</u> Firm-years with missing values of variables used in the model for measuring unexpected operating cash flows	203	22,130
	1,766	20,493
<u>Less:</u> Firm-years with missing values of investing or financing cash flows	31	766
	1,735	19,727
<u>Less:</u> Firm-years with missing values of Altman's (2002) Z-Score	88	2,715
	1,647	17,012
<u>Less:</u> Firm-years with missing values of control variables	29	1,347
	1,618	15,665
<u>Less:</u> Firm-years in industry-years with observations less than 10 (minimum requirement for running industry-year regressions for estimating unexpected operating cash flows)	145	1,896
	1,473	13,769
<u>Less:</u> Firm-years in financial services industries (NIC Codes: 64, 65 and 66)	103	464
Final sample (March 1995-March 2011)	1,370	13,305

Table 3
Descriptive statistics

	N	Mean	p25	Median	p75	Std Dev
sale	13305	6030.256	300.800	974.400	3377.500	22983.750
at	13305	7741.137	334.900	1077.300	3825.500	30976.737
ocf	13305	629.438	4.000	47.300	234.100	3073.784
icf	13305	-634.496	-236.000	-41.200	-4.900	2938.512
fcf	13305	150.554	-58.300	-3.200	58.400	1850.756
cfo	13305	0.071	0.013	0.074	0.138	0.129
def_sale	13305	1.213	0.676	1.028	1.496	0.860
ch_sale	13305	0.184	0.006	0.116	0.288	0.367
ue_cfo	13305	-0.001	-0.067	-0.001	0.068	0.151
ivo	13305	0.109	0.013	0.055	0.136	0.169
ivi	13305	0.005	0.000	0.000	0.000	0.022
fino	13305	0.038	0.000	0.009	0.060	0.054
fini	13305	0.083	0.000	0.000	0.072	0.211
cff	13305	0.045	-0.060	-0.009	0.072	0.232
cfi	13305	-0.104	-0.136	-0.055	-0.013	0.174
roa	13305	0.063	0.015	0.047	0.098	0.088
size	13305	7.138	5.814	6.982	8.249	1.742
mtb	13305	1.118	0.229	0.575	1.258	1.772
dacc	13305	0.002	-0.061	-0.002	0.057	0.127
zscore	13305	5.097	3.332	4.470	5.964	3.134

All variables are winsorized at 1% and 99% by fiscal year. Variable definitions are given in Table 1.

Table 4
Pearson (Spearman) correlations above (below) the diagonal

variable	sale	at	cfo	def_sale	ch_sale	ue_cfo	ivo	ivi
sale	1	0.861***	0.063***	0.100***	0.061***	0.045***	0	0.006
at	0.914***	1	0.040***	-0.037***	-0.009	0.036***	0.022**	0
cfo	0.152***	0.096***	1	-0.005	-0.091***	0.709***	0.093***	-0.197***
def_sale	0.250***	-0.070***	0.098***	1	0.632***	-0.015*	0.106***	-0.056***
ch_sale	0.205***	0.050***	0.039***	0.579***	1	-0.006	0.229***	-0.030***
ue_cfo	0.041***	0.016*	0.767***	0.021**	0.039***	1	0.110***	-0.135***
ivo	0.162***	0.157***	0.286***	0.184***	0.272***	0.207***	1	-0.156***
ivi	-0.088***	-0.035***	-0.227***	-0.161***	-0.142***	-0.160***	-0.593***	1
fino	-0.002	-0.048***	0.487***	-0.042***	-0.163***	0.375***	-0.405***	0.166***
fini	0.049***	0.085***	-0.396***	0.118***	0.246***	-0.310***	0.505***	-0.182***
cff	0.020**	0.070***	-0.488***	0.075***	0.212***	-0.377***	0.485***	-0.181***
cfi	-0.161***	-0.156***	-0.288***	-0.183***	-0.271***	-0.210***	-0.999***	0.598***
roa	0.288***	0.220***	0.303***	0.355***	0.424***	0.215***	0.395***	-0.147***
size	0.914***	1.000***	0.096***	-0.070***	0.050***	0.016*	0.157***	-0.035***
mtb	0.236***	0.242***	0.054***	0.077***	0.167***	0.083***	0.182***	-0.043***
dacc	0.017*	0.041***	-0.523***	0.012	-0.010	-0.602***	-0.018**	0.071***
zscore	0.121***	0.129***	0.007	-0.061***	0.002	-0.023***	-0.030***	-0.047***

All variables are winsorized at 1% and 99% by fiscal year. Variable definitions are given in Table 1.

* p<0.10, ** p<0.05, *** p<0.01

Table 4 (Contd.)
Pearson (Spearman) Correlations above (below) the diagonal

variable	fino	fini	cff	cfi	roa	size	mtb	dacc	zscore
sale	-0.007	-0.026***	-0.022**	0.001	0.107***	0.507***	0.160***	-0.019**	0.041***
at	-0.037***	-0.007	0.003	-0.022**	0.080***	0.523***	0.127***	-0.004	0.031***
cfo	0.415***	-0.359***	-0.424***	-0.115***	0.220***	0.085***	0.055***	-0.579***	0.015*
def_sale	-0.011	0.150***	0.139***	-0.110***	0.278***	-0.076***	0.047***	0.025***	0.036***
ch_sale	-0.090***	0.302***	0.296***	-0.227***	0.370***	0.007	0.105***	0.029***	-0.003
ue_cfo	0.298***	-0.212***	-0.262***	-0.125***	0.180***	0.018**	0.085***	-0.601***	-0.018**
ivo	-0.255***	0.757***	0.749***	-0.992***	0.345***	0.098***	0.117***	0.037***	-0.069***
ivi	0.143***	-0.064***	-0.092***	0.278***	0.013	-0.016*	0.022**	0.142***	-0.112***
fino	1	-0.271***	-0.481***	0.266***	0.049***	-0.039***	0.021**	-0.267***	0.059***
fini	-0.852***	1	0.974***	-0.744***	0.260***	0.051***	0.100***	0.288***	-0.056***
cff	-0.952***	0.914***	1	-0.740***	0.226***	0.056***	0.086***	0.325***	-0.065***
cfi	0.406***	-0.505***	-0.485***	1	-0.334***	-0.097***	-0.111***	-0.018**	0.053***
roa	-0.024***	0.133***	0.084***	-0.390***	1	0.185***	0.260***	0.254***	-0.257***
size	-0.048***	0.085***	0.070***	-0.156***	0.220***	1	0.230***	0.022***	0.102***
mtb	-0.073***	0.126***	0.109***	-0.180***	0.318***	0.242***	1	-0.008	0.011
dacc	-0.324***	0.306***	0.348***	0.023***	0.209***	0.041***	-0.004	1	-0.089***
zscore	0.019**	0.009	-0.023***	0.026***	-0.297***	0.129***	-0.006	-0.116***	1

All variables are winsorized at 1% and 99% by fiscal year. Variable definitions are given in Table 1. * p<0.10, ** p<0.05, *** p<0.01

Table 5
Model fit statistics (Dechow et al. 1998)

	Predicted Sign	Dependent Variable: <i>def_cfo</i>
inv_at _{t-1}	-	-5.697*** (-4.341)
def_sale _t	+	0.046*** (4.570)
ch_sale _t	-	-0.044* (-1.833)
Constant		0.042*** (6.815)
Adjusted R-Square		16.0%
No. of Industry-Years		376
No. of Observations		13305

The regression model is $CFO_{i,t}/A_{i,t-1} = \beta_0 + \beta_1(1/A_{i,t-1}) + \beta_2(S_{i,t}/A_{i,t-1}) + \beta_3(\Delta S_{i,t}/A_{i,t-1}) + \varepsilon_{i,t}$. Amounts reported are means of coefficients from industry-year regressions using model of Dechow et al. (1998). t-statistics are given in parentheses. Variable definitions are given in Table 1. * p<0.10, ** p<0.05, *** p<0.01

Table 6
Regressions of unexpected operating cash flows on
investing cash outflows, investing cash inflows, financing
cash outflows, and financing cash inflows

	Predicted Sign	Dependent Variable: <i>ue_cfo</i>	
		Panel A	Panel B
<i>ivo_t</i>	+	0.295*** (19.214)	0.331*** (21.265)
<i>ivi_t</i>	-	-0.408*** (-5.868)	-0.463*** (-6.696)
<i>fino_t</i>	+	0.410*** (11.556)	0.407*** (11.444)
<i>fini_t</i>	-	-0.250*** (-21.501)	-0.273*** (-23.100)
<i>zscore_{t-1}</i>	?	-0.000 (-0.696)	-0.000 (-0.025)
<i>zscore_{t-1}*ivo_t</i>	+	0.004** (2.052)	0.005*** (2.715)
<i>zscore_{t-1}*ivi_t</i>	-	-0.005 (-0.659)	-0.007 (-0.926)
<i>zscore_{t-1}*fino_t</i>	+	0.009 (1.464)	0.008 (1.309)
<i>zscore_{t-1}*fini_t</i>	-	-0.002* (-1.790)	-0.003** (-2.412)
<i>roa_t</i>	+	0.514*** (40.789)	0.550*** (42.913)
<i>size_t</i>	-	-0.003*** (-5.669)	-0.003*** (-4.280)
<i>mtb_t</i>	+	0.001 (0.986)	0.001* (1.931)
<i>dacc_t</i>	-	-0.653*** (-76.284)	-0.656*** (-75.130)
Constant		-0.034*** (-7.252)	-0.045*** (-5.277)
Industry dummies		No	Yes
Year dummies		No	Yes
No. of observations		13305	13305
Adjusted R-Square		53.3%	54.7%
p-value		0.000	0.000

The regression model is $UE_CFO_{i,t} = \alpha_0 + \alpha_1 IVO_{i,t} + \alpha_2 IVI_{i,t} + \alpha_3 FINO_{i,t} + \alpha_4 FINI_{i,t} + \alpha_5 ROA_{i,t} + \alpha_6 SIZE_{i,t} + \alpha_7 MTB_{i,t} + \alpha_8 DACC_{i,t} + \alpha_9 ZSCORE_{i,t-1} + \alpha_{10} ZSCORE_{i,t-1} * IVO_{i,t} + \alpha_{11} ZSCORE_{i,t-1} * IVI_{i,t} + \alpha_{12} ZSCORE_{i,t-1} * FINO_{i,t} + \alpha_{13} ZSCORE_{i,t-1} * FINI_{i,t} + \omega_{i,t}$. Amounts reported are coefficients from pooled regressions. t-statistics are given in parentheses. Variable definitions are given in Table 1. * p<0.10, ** p<0.05, *** p<0.01

Table 7
Regressions of unexpected operating cash flows on net investing and financing cash flows

	Predicted Sign	Dependent Variable: <i>ue_cfo</i>	
		Panel A	Panel B
cff_t	-	-0.275*** (-25.891)	-0.293*** (-26.716)
cfi_t	-	-0.305*** (-21.124)	-0.340*** (-23.047)
$zscore_{t-1}$	?	0.000 (0.570)	0.000 (1.008)
$zscore_{t-1} * cff_t$	-	-0.004*** (-2.941)	-0.004*** (-3.372)
$zscore_{t-1} * cfi_t$	-	-0.004** (-2.362)	-0.005*** (-3.031)
roa_t	+	0.527*** (42.757)	0.563*** (45.003)
$size_t$	-	-0.003*** (-6.144)	-0.003*** (-4.495)
mtb_t	+	0.001 (1.041)	0.001** (2.163)
$dacc_t$	-	-0.662*** (-78.092)	-0.665*** (-77.336)
Constant		-0.026*** (-6.048)	-0.039*** (-4.679)
Industry dummies		No	Yes
Year dummies		No	Yes
No. of observations		13305	13305
Adjusted R-Square		53.1%	54.6%
p-value		0.000	0.000

The regression model is $UE_CFO_{i,t} = \alpha_0 + \alpha_1 CFF_{i,t} + \alpha_2 CFI_{i,t} + \alpha_3 ROA_{i,t} + \alpha_4 SIZE_{i,t} + \alpha_5 MTB_{i,t} + \alpha_6 DACC_{i,t} + \alpha_7 ZSCORE_{i,t-1} + \alpha_8 ZSCORE_{i,t-1} * CFF_{i,t} + \alpha_9 ZSCORE_{i,t-1} * CFI_{i,t} + \delta_{i,t}$. Amounts reported are coefficients from pooled regressions. t-statistics are given in parentheses. Variable definitions are given in Table 1. * p<0.10, ** p<0.05, *** p<0.01

Table 8
Regressions of unexpected operating cash flows on
investing cash outflows, investing cash inflows,
financing cash outflows, and financing cash inflows
(Comparison of firms in India and the United States)

	Predicted Sign	Dependent Variable: <i>ue_cfo</i>	
		Panel A	Panel B
ivo_t	+	0.302*** (80.083)	0.315*** (83.652)
ivi_t	-	-0.424*** (-44.099)	-0.513*** (-53.917)
fin_t	+	0.614*** (59.373)	0.651*** (64.040)
$fini_t$	-	-0.228*** (-89.824)	-0.240*** (-95.747)
$zscore_{t-1}$	?	-0.000 (-1.117)	0.000*** (4.305)
$zscore_{t-1} * ivo_t$	+	-0.002*** (-14.201)	-0.001*** (-12.111)
$zscore_{t-1} * ivi_t$	-	-0.005*** (-10.996)	-0.006*** (-11.870)
$zscore_{t-1} * fino_t$	+	0.007*** (7.444)	0.007*** (7.941)
$zscore_{t-1} * fini_t$	-	0.001*** (14.464)	0.001*** (12.195)
ind_t	?	-0.032*** (-16.278)	-0.022 (-0.000)
$ind_t * ivo_t$	+	0.302*** (16.338)	0.363*** (19.574)
$ind_t * ivi_t$	-	-0.532*** (-5.929)	-0.575*** (-6.530)
$ind_t * fino_t$	+	0.241*** (5.648)	0.291*** (6.873)
$ind_t * fini_t$	-	-0.277*** (-18.803)	-0.316*** (-21.677)
$ind_t * zscore_{t-1} * ivo_t$	+	0.012*** (5.192)	0.016*** (6.933)
$ind_t * zscore_{t-1} * ivi_t$	-	-0.010 (-0.991)	-0.011 (-1.139)
$ind_t * zscore_{t-1} * fino_t$	+	0.003 (0.427)	0.011 (1.487)
$ind_t * zscore_{t-1} * fini_t$	-	-0.010*** (-5.533)	-0.012*** (-6.920)

roa _t	+	0.191*** (84.812)	0.213*** (94.556)
size _t	-	-0.006*** (-27.120)	-0.006*** (-26.743)
mtb _t	+	0.002*** (16.028)	0.001*** (9.941)
dacc _t	-	-0.008*** (-13.715)	-0.010*** (-17.248)
Constant		0.012*** (9.007)	-0.024 (-1.247)
Industry dummies		No	Yes
Year dummies		No	Yes
No. of observations		89563	89563
Adjusted R-Square		37.6%	40.7%
p-value		0.000	0.000

The regression model is $UE_CFO_{i,t} = \alpha_0 + \alpha_1 IVO_{i,t} + \alpha_2 IVI_{i,t} + \alpha_3 FINO_{i,t} + \alpha_4 FINI_{i,t} + \alpha_5 ZSCORE_{i,t-1} + \alpha_6 ZSCORE_{i,t-1} * IVO_{i,t} + \alpha_7 ZSCORE_{i,t-1} * IVI_{i,t} + \alpha_8 ZSCORE_{i,t-1} * FINO_{i,t} + \alpha_9 ZSCORE_{i,t-1} * FINI_{i,t} + \alpha_{10} IND_{i,t} + \alpha_{11} IND_{i,t} * IVO_{i,t} + \alpha_{12} IND_{i,t} * IVI_{i,t} + \alpha_{13} IND_{i,t} * FINO_{i,t} + \alpha_{14} IND_{i,t} * FINI_{i,t} + \alpha_{15} IND_{i,t} * ZSCORE_{i,t-1} * IVO_{i,t} + \alpha_{16} IND_{i,t} * ZSCORE_{i,t-1} * IVI_{i,t} + \alpha_{17} IND_{i,t} * ZSCORE_{i,t-1} * FINO_{i,t} + \alpha_{18} IND_{i,t} * ZSCORE_{i,t-1} * FINI_{i,t} + \alpha_{19} ROA_{i,t} + \alpha_{20} SIZE_{i,t} + \alpha_{21} MTB_{i,t} + \alpha_{22} DACC_{i,t} + \mu_{i,t}$. Amounts reported are coefficients from pooled regressions. t-statistics are given in parentheses. Variable definitions are given in Table 1. * p<0.10, ** p<0.05, *** p<0.01

Table 9
Regressions of unexpected operating cash flows on net
investing and financing cash flows
(Comparison of firms in India and the United States)

	Predicted Sign	Dependent Variable: <i>ue_cfo</i>	
		Panel A	Panel B
cff_t	-	-0.252*** (-104.346)	-0.268*** (-112.067)
cfi_t	-	-0.314*** (-92.131)	-0.339*** (-99.514)
$zscore_{t-1}$	?	-0.000 (-1.617)	0.000*** (4.624)
$zscore_{t-1} * cff_t$	-	0.001*** (11.522)	0.000*** (9.184)
$zscore_{t-1} * cfi_t$	-	0.001*** (10.671)	0.001*** (8.089)
ind_t	?	-0.033*** (-20.248)	0.041 (0.000)
$ind_t * cff_t$	-	-0.332*** (-25.401)	-0.369*** (-28.174)
$ind_t * cfi_t$	-	-0.342*** (-19.884)	-0.395*** (-22.585)
$ind_t * zscore_{t-1} * cff_t$	-	-0.013*** (-7.879)	-0.015*** (-9.478)
$ind_t * zscore_{t-1} * cfi_t$	-	-0.013*** (-6.291)	-0.017*** (-8.044)
roa_t	+	0.191*** (84.165)	0.212*** (93.449)
$size_t$	-	-0.006*** (-26.392)	-0.006*** (-25.820)
mtb_t	+	0.002*** (17.616)	0.001*** (11.826)
$dacc_t$	-	-0.008*** (-13.719)	-0.010*** (-17.158)
Constant		0.022*** (16.883)	-0.014 (-0.718)
Industry dummies		No	Yes
Year dummies		No	Yes
No. of observations		89563	89563
Adjusted R-Square		36.4%	39.4%
p-value		0.000	0.000

The regression model is $UE_CFO_{i,t} = \alpha_0 + \alpha_1 CFF_{i,t} + \alpha_2 CFI_{i,t} + \alpha_3 ZSCORE_{i,t-1} + \alpha_4 ZSCORE_{i,t-1} * CFF_{i,t} + \alpha_5 ZSCORE_{i,t-1} * CFI_{i,t} + \alpha_6 IND_{i,t} + \alpha_7 IND_{i,t} * CFF_{i,t} + \alpha_8 IND_{i,t} * CFI_{i,t} + \alpha_9 IND_{i,t} * ZSCORE_{i,t-1} * CFF_{i,t} + \alpha_{10} IND_{i,t} * ZSCORE_{i,t-1} * CFI_{i,t} + \alpha_{11} ROA_{i,t} + \alpha_{12} SIZE_{i,t} + \alpha_{13} MTB_{i,t} + \alpha_{14} DACC_{i,t} + \Omega_{i,t}$. Amounts reported are coefficients from pooled

regressions. t-statistics are given in parentheses. Variable definitions are given in Table 1.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$